Counting the right interviews

Two overlapping extracts, three survey waves and one decision that governs every result: who was actually in a position to answer?

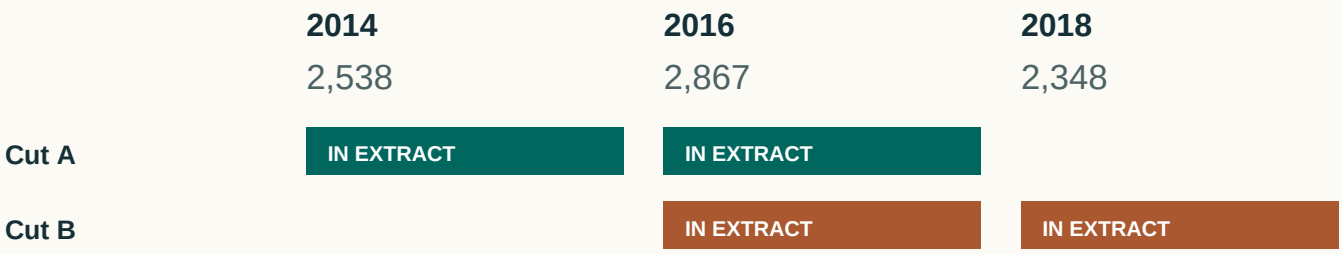
7,753

distinct survey interviews

One row per survey year and respondent ID. These are repeated cross sections, not a panel following the same people.

Where the two extracts meet

INTERVIEWS BY SURVEY WAVE



A has 5,405 rows; B has 5,215. All 2,867 shared interviews agree field for field (53.04% of A; 54.98% of B). ID alone would collapse 4,886 separate interviews. Membership remains available for audit.

Why these years?

I chose 2014/2016 and 2016/2018. Both cuts are near 5,000 interviews, every requested concept is observed, and 2016 tests whether overlapping answers agree. The earlier 2012-2016 option yielded fewer web-duration observations; including 2021 introduced mode and wording changes.

TRADEOFF / COMPARABILITY OVER RECENCY

These years support the same field definitions across adjacent waves. They do not describe today's audience. Use newer compatible data or a direct audience study for a current decision.

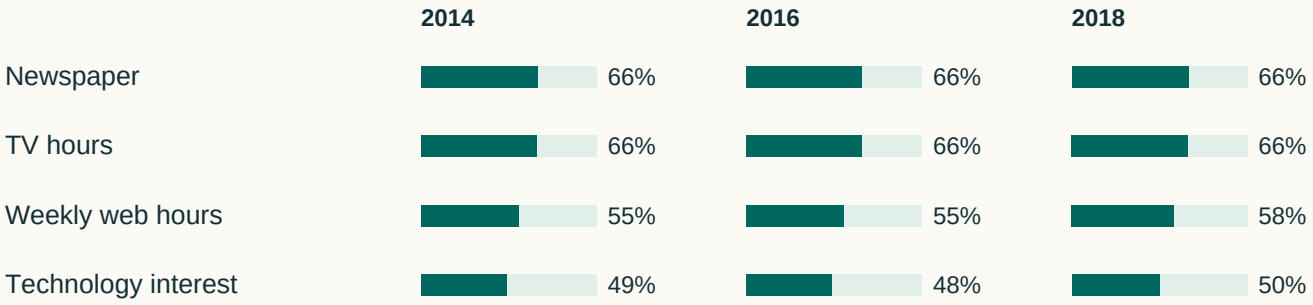
READING PATH Use the tabs above. On Research, switch the survey weight to inspect the estimate.

Who could answer?

The share missing varies by question. Start with the intended population, then apply only the filters that question needs.

Valid responses by wave

Bars show share of all interviews; each track is 0-100%.



Ballot C had no valid news, TV, computer-use or web-hours answers in any wave. INTTECH had valid answers on B/X, B/Y and C/Y in 2014, then A/X, A/Y and B/X in 2016 and 2018. Letters before the slash identify the ballot and letters after it identify the form. Of 3,797 who answered, 41% were very interested and 46% moderately interested.

Question map / fields and eligibility

Work and history

WRKSTAT includes temporary absence as working. EVWORK asks about at least one year. Current workers and job seekers skipped it, so their lifetime duration is unknown. Q5 bounds the missing answers.

Education and industry

DEGREE measures credentials; EDUC measures years. Use INDUS10 for current workers so a prior industry's label is not mistaken for a current job.

Satisfaction and media

In 2016, Ballot A/Form Y used alternate work-satisfaction wording. Q4 keeps only standard SATJOB. TVHOURS measures daily duration, NEWS measures frequency, and zero hours counts.

Faith and technology

RELIG16 is upbringing, RELIG current preference, ATTEND attendance, RELITEN strength. PRAY is ordered frequency; POSTLIFE is belief. INTTECH is interest, not usage.

Web duration

WWWHR + WWWMIN / 60 measures weekly non-email web time. Require both fields. A skip is unknown. Age, sex, region, ballot, form and mode retain context.

Pre-build profile: docs/profile.csv records per-variable, per-year missing shares. All mapped concepts have observations in all three waves.

What the six analyses show

Each result counts the shared wave once. Percentages on this page describe observed respondents and are unweighted.

01 Prayer and upbringing

Several-times-daily prayer: 1,413/4,007 raised Protestant, 601/2,515 raised Catholic and 28/49 raised Muslim. The last group is small. CSV 01 includes the full distribution.

02 Belief in life after death

Yes among valid answers: 2014: 79.53% (1,783/2,242), 2016: 80.72% (2,089/2,588), 2018: 81.02% (1,720/2,123). Counts coded not asked that year were 0, 0, 0, respectively. Uncertain and unanswered items are excluded.

03 Leading industry by degree

Among current workers with both answers: associate/junior college: hospitals (41/398); bachelor's: elementary and secondary schools (109/1,037); graduate: elementary and secondary schools (94/582); high school: construction (186/2,275); less than high school: construction (53/384). Each is the most frequent industry, not necessarily a majority.

04 TV hours and job satisfaction

Very satisfied	Moderately satisfied	A little dissatisfied	Very dissatisfied
2.32	**2.37**	**2.39**	**2.55**
hours/day; n=1,415	hours/day; n=1,091	hours/day; n=251	hours/day; n=88

Of 4,712 workers, 3,911/4,423 who answered were satisfied (88.42%); among TV pairs: 2,506/2,845 (88.08%). Median TV: 2 h in each group. Neither test detected an association: Spearman rho -0.011, p=0.57; Kruskal-Wallis p=0.44. Tests ignore survey design.

05 Ever worked at least one year

Of 3,034 nonworkers, 306 job seekers were skipped and 2 eligible answers missing. Yes: 2,467/2,726 (90.5%). Across all nonworkers the possible range is 81.31-91.46%.

06 News and TV across waves

Daily readership: 24.9% / 21.1% / 20.0%. Mean TV hours: 2.98 / 3.03 / 2.94. Only the first news drop exceeded two approximate standard errors. These are separate samples; the comparison does not establish a national trend.

Q7 / See Sample for the overlap audit. Detailed counts: report/results/01-07.

What I built, and why

One local batch, one interview key, and an explicit stopping point when a file, model or test fails.

From immutable file to usable answer

01 RAW	02 MEMBERSHIP	03 ANSWERS	04 RESPONDENTS
File + row	Year + ID	Item + status	One interview

Python reads the strict JSON export (the companion is a Python literal), validates aligned arrays and keys, and loads DuckDB in a transaction. dbt tables preserve membership, decode each variable once, then pivot answers. Versioned seeds distinguish genuine zero from inapplicable, don't know, no answer and not asked that year. Lower snake case; _value is numeric, _status explains availability.

Three engineering decisions

MATERIALIZE / PAY ONCE

A view re-parsed raw keys on each overlap audit. Materializing the four keys adds build cost and storage but avoids repeated parsing. The slowest baseline gave identical results.

RERUN / VALIDATE AGAIN

Revalidate unchanged files against current metadata; skip duplicate inserts. A conflict rolls back the raw load. dbt build and tests gate docs and staged CSV publication. CI runs on pushes and PRs; failures stop publication.

SCALE / CHANGE ONLY WHEN NEEDED

DuckDB avoids a server for two periodic extracts. At full scale the dense model approaches 530 million cells. Read only needed fields and rebuild affected waves before adding managed compute.

Overlap query / measured runtime

Median of three warm runs, one thread. New extracts are inserted once; downstream tables rebuild.

Oversight and effort: I spent roughly six hours choosing extracts, discussing tradeoffs and reviewing the work. I supplied both cuts and challenged the TV and employment interpretations. Codex wrote the code and SQL, ran checks and helped write the report; an AI reviewer challenged the scope. For population claims I would next add survey strata and sampling units.

Two questions for Yaqeen

The survey can motivate a research question. A decision about Yaqeen's readers still needs evidence from those readers.

01 Does afterlife belief differ by prayer frequency?

Switch weight: WTSSPS | WTSSNRPS

WTSSPS / primary

	2014	2016	2018
At least daily	**89.2%** n=1,354	**90.4%** n=1,576	**89.1%** n=1,286
Less than daily	**65.0%** n=880	**64.3%** n=995	**68.4%** n=832

Both weights give the same direction in every wave. For each group, divide the weights of yes answers by the weights of valid yes/no answers. Daily prayer and afterlife belief are associated; neither question measures Muslim conviction or a causal effect.

02 Does media use differ by age?

2018 / same respondents answered both items

AGE	PAIRED / KNOWN AGE	DAILY NEWS	WEB HOURS / WEEK*
18-34	376 / 632	8.0%	**14**
35-54	479 / 771	17.1%	**10**
55+	500 / 938	30.6%	**4**

*Median weekly non-email web hours. Axes run to 100% and 16 hours. Ballot C and missing pairs are excluded. Counting explicit nonusers with routed skips as zero adds 57 interviews in 2018. The resulting age-group medians are 14, 10, 4 hours. Web time alone cannot establish access or format preference.

NEXT DECISION / TEST A FORMAT, NOT A FAITH PROFILE

Offer a paper and video on one topic to consenting readers. Compare completion and learning separately. Historical US respondents are not Yaqeen's audience; don't infer preferences or target individuals by religion.

If scale requires AWS: versioned S3, scheduled ECS jobs and Redshift; separate roles and track compute and scan cost. Compare Pew at survey level and keep pilot aggregates separate. See docs/production.md for assumptions.